

COMP-SCI-5565

Intro to Statistical Learning

Topic 4 – Resampling Methods

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Lecture Objectives

- Use a validation set to estimate the test error of a predictive model.
- Use leave-one-out cross-validation to estimate the test error of a predictive model.
- Use K-fold cross-validation to estimate the test error of a predictive model.
- Use the bootstrap to obtain standard errors of an estimate.
- Describe the advantages and disadvantages of the various methods for estimating model test error.

Outline

4.1 Overview of Resampling Methods

4.2 Validation-Set Approach

4.3 Leave-One-Out Cross-Validation

4.4 K-Fold Cross-Validation

4.5 Bootstrap

Outline

4.1 Overview of Resampling Methods

4.2 Validation-Set Approach

4.3 Leave-One-Out Cross-Validation

4.4 K-Fold Cross-Validation

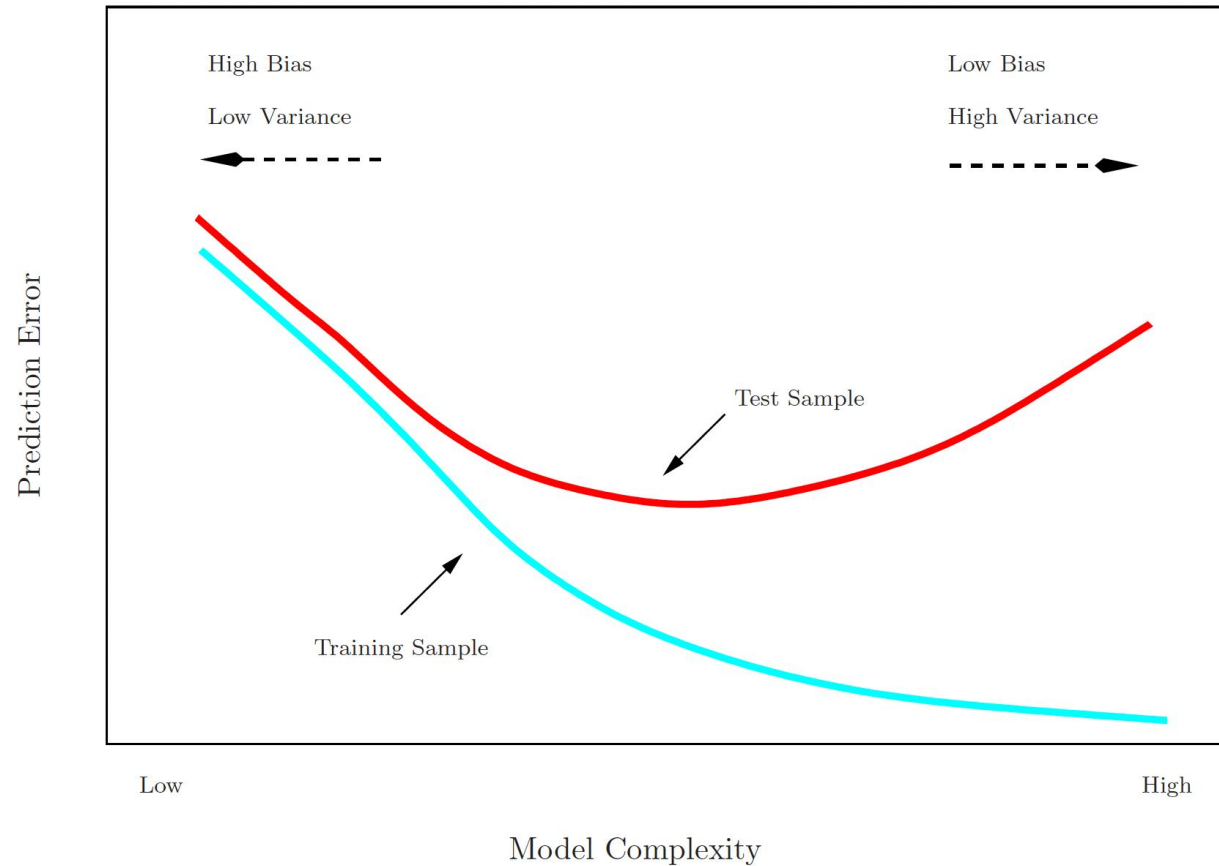
4.5 Bootstrap

4.1 Overview of Resampling Methods

- We discuss two resampling methods: cross-validation and bootstrap.
- Recall the distinction between the **test error** and the **training error**.
- The **test error** is the average error resulting from using a statistical learning method to predict the response on a new observation, which was not used in training the method.
- The **training error** can be easily calculated by applying the statistical learning method to the observations used in its training.
- However, the training error rate is often quite different from the test error rate, and in particular, the former can dramatically underestimate the latter.

4.1 Overview of Resampling Methods

Training- versus Test-Set Performance



4.1 Overview of Resampling Methods

- The best solution is to have a large test set (which is not often available).
- Some methods (AIC, BIC) make a mathematical adjustment to the training error rate to estimate the test error rate.
- Here, we consider methods that estimate the test error by **holding out** a subset of the training observations from the fitting (training) process.
- Then testing the statistical learning method to those held-out observations.

Outline

4.1 Overview of Resampling Methods

4.2 Validation-Set Approach

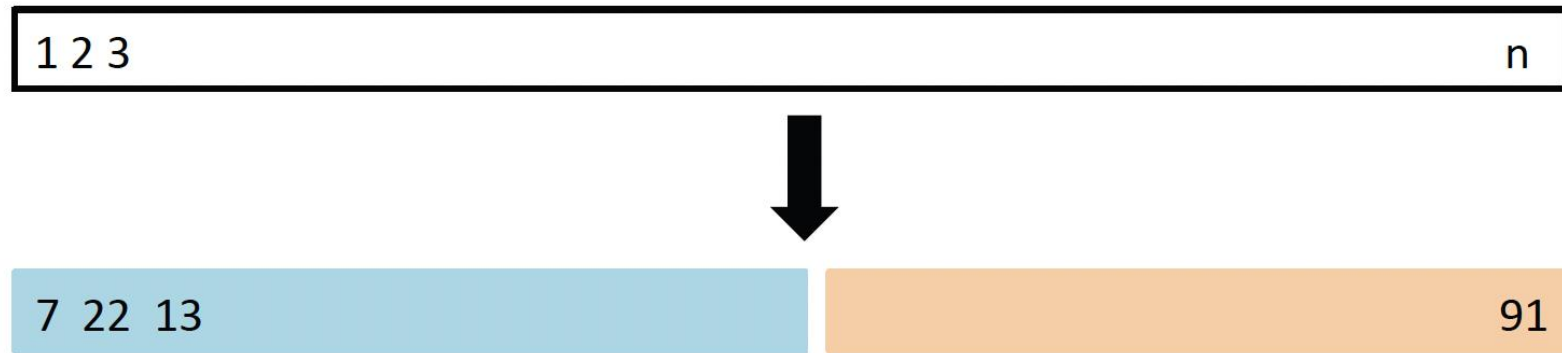
4.3 Leave-One-Out Cross-Validation

4.4 K-Fold Cross-Validation

4.5 Bootstrap

4.2 Validation-Set Approach

- This involves randomly splitting the data into training and validation sets.
- Note that in certain applications, such as time series analysis, it is not feasible to split the data randomly.



A random splitting into two halves: the left part is the training set, the right part is the validation set.

4.2 Validation-Set Approach

Advantages

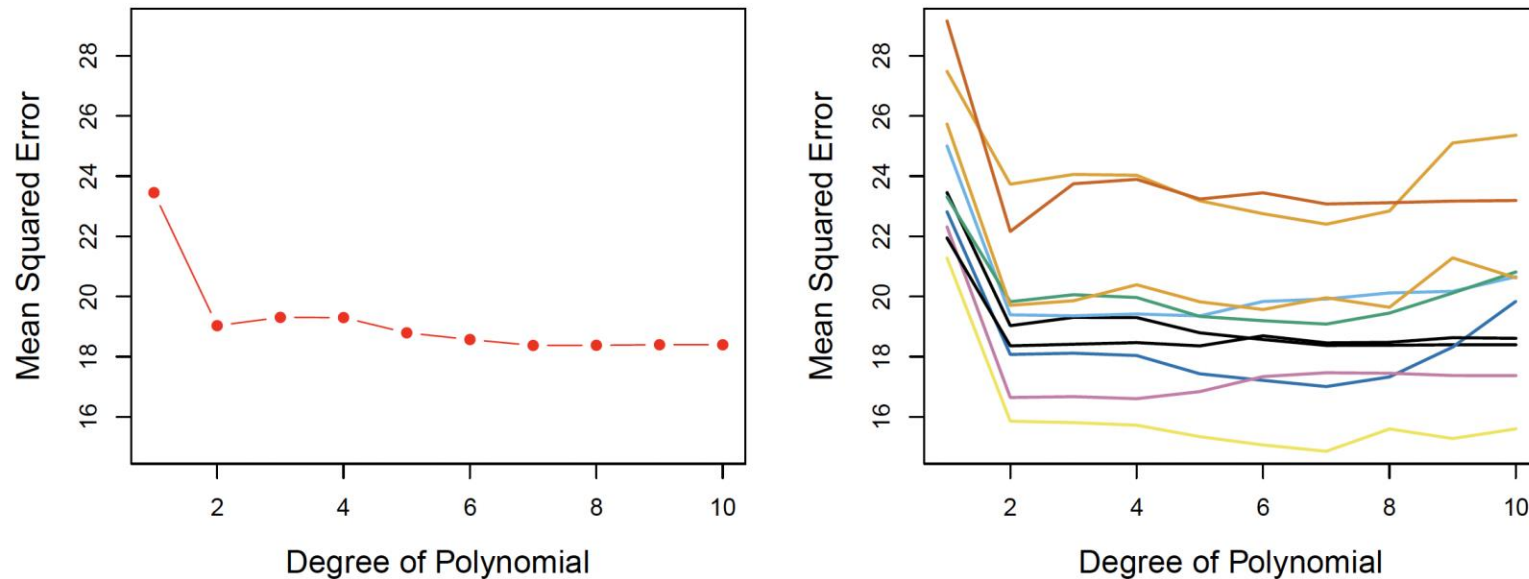
- The advantage of the validation set approach is that it is conceptually simple to understand and implement.

Disadvantages

- The validation error rate varies depending on the training and validation sets assigned.
- We give up valuable data points by not using all the data to estimate the model.
- Thus, the validation error rate will likely overestimate the test error rate.

4.2 Validation-Set Approach

Validation Error Rate Varies Depending on the Data Set



Left: Validation error estimates for a single split into training and validation data sets. Right: The validation method was repeated ten times, each time using a different random split of the observations into a training set and a validation set. This illustrates the variability in the estimated test MSE that results from this approach.

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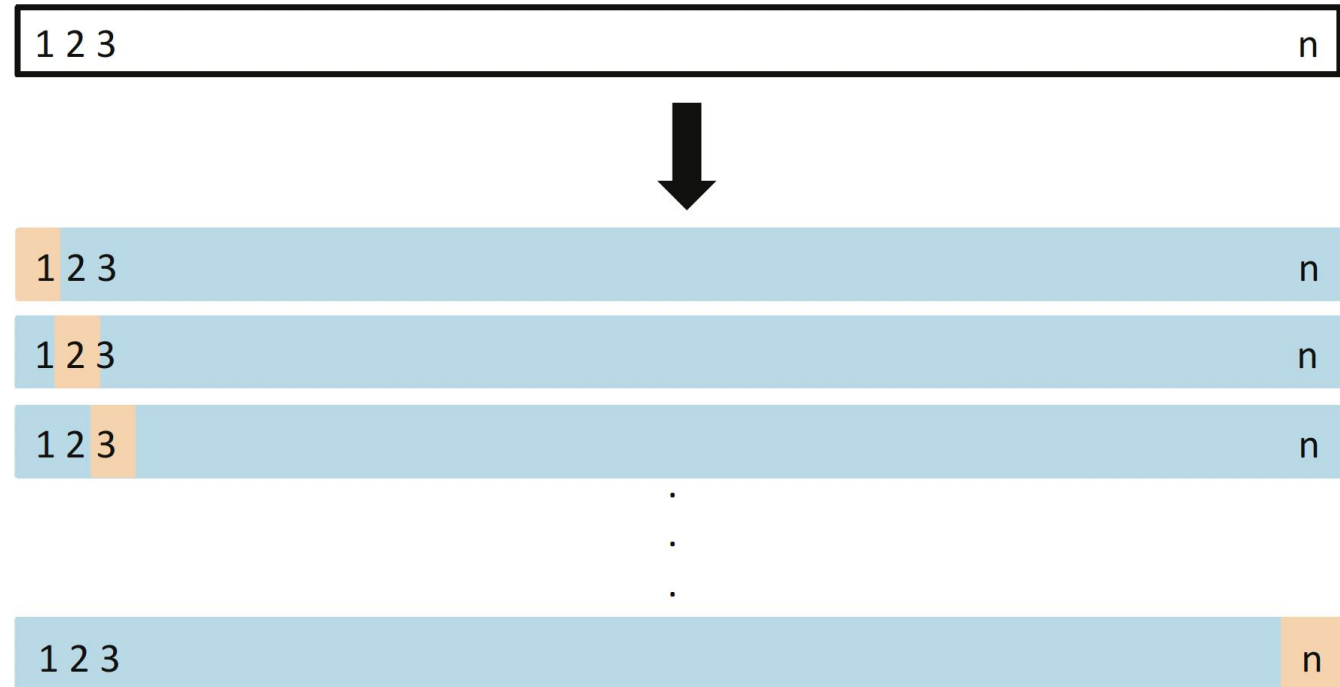
4.5 Bootstrap

4.3 Leave-One-Out Cross-Validation (LOOCV)

- LOOCV aims to address some of the drawbacks of the validation set approach.
- Like the validation set approach, LOOCV splits the data into training and validation sets.
- However, the validation set includes one observation, and the training set includes $n - 1$ observations.
- This process is repeated for all observations such that n models are estimated.
- LOOCV estimate of test error is averaged over the n models:

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n MSE_i$$

4.3 Leave-One-Out Cross-Validation (LOOCV)



A schematic display of LOOCV. A set of n data points is repeatedly split into a training set (shown in blue) containing all but one observation and a validation set that contains only that observation (shown in beige).

4.3 Leave-One-Out Cross-Validation (LOOCV)

Advantages

- It has less bias since models are repeatedly fitted on slightly different data sets, so it tends not to overestimate the test error as much as the validation set approach.
- The estimated test error will always be the same when LOOCV is performed on the entire data set.

Disadvantages

- The major disadvantage of LOOCV is that it is computationally expensive.

LOOCV can generally be used for various models, including logistic regression, LDA, Naïve Bayes, and KNN.

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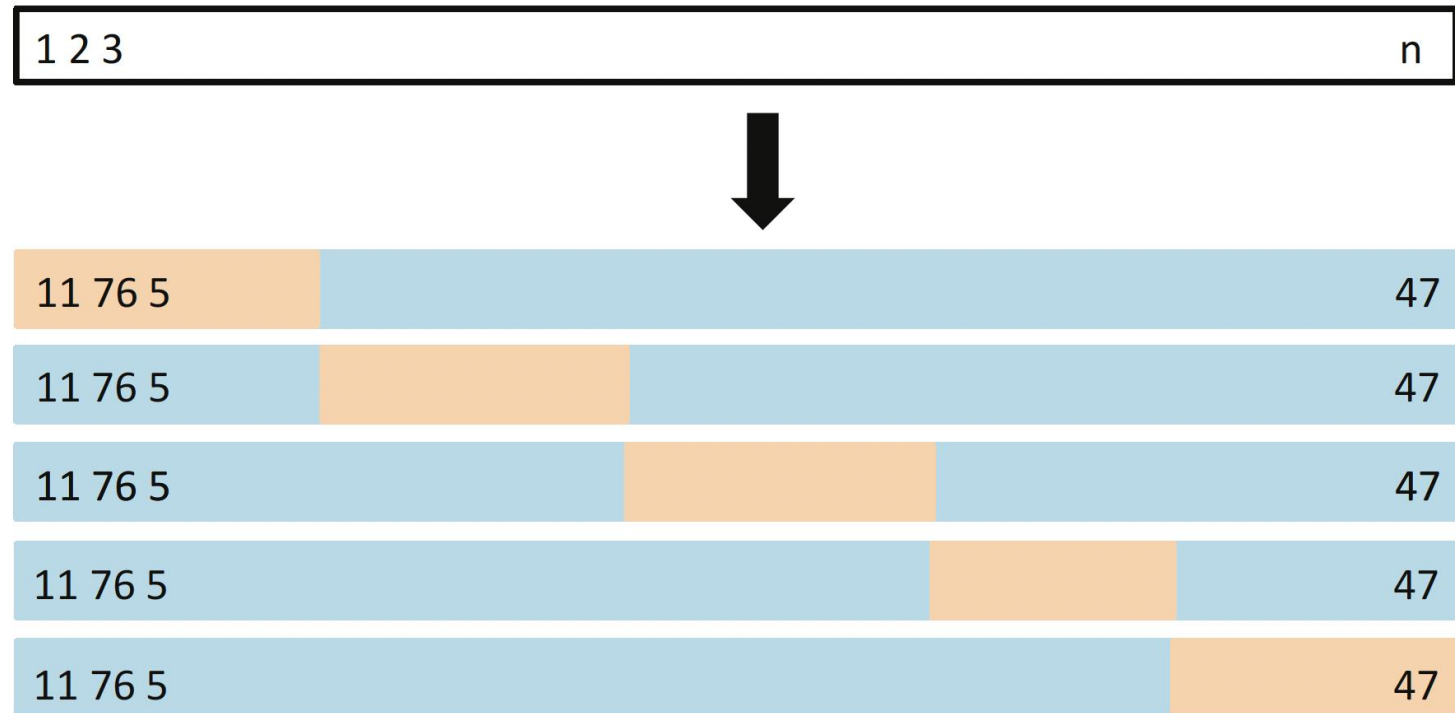
4.4 K-Fold Cross-Validation

4.5 Bootstrap

4.4 K-Fold Cross-Validation

- This is an alternative to LOOCV, which involves dividing the data set into k groups (folds) of approximately equal size.
- The idea is to randomly divide the data into K equal-sized parts.
- We leave out part k , fit the model to the other $K - 1$ parts (combined).
- Then, we test the model using the left-out k th part.
- This is done in turn for each part $k = 1, 2, \dots, K$, and then the results are combined.
- The percent of the data set that is in the validation set can be thought of as $1/K$.
- E.g., for $k=5$ groups, 20% of the data would be withheld for testing.

4.4 K-Fold Cross-Validation



A schematic display of a 5-fold CV, $K = 5$. A set of n observations is randomly split into five non-overlapping groups. Each fifth acts as a validation set (shown in beige), and the remainder as a training set (shown in blue). The test error is estimated by averaging the five resulting MSE estimates.

4.4 K-Fold Cross-Validation

- Thus, LOOCV is a special case of k-fold cross-validation, where $k = n$.
- The equation for the CV statistic is below:

$$CV_k = \frac{1}{k} \sum_{i=1}^k MSE_i$$

Advantages

- The main advantage of k-fold over LOOCV is computational.
- However, there are other advantages related to the bias-variance tradeoff.

4.4 K-Fold Cross-Validation

Bias-variance tradeoff and K-fold cross-validation

- The validation approach tends to overestimate the true test error, but there is low variance in the estimate since we have one estimate of the test error.
- Conversely, the LOOCV method has little bias since almost all observations are used to create the models.
- However, LOOCV doesn't shake up the data enough: the estimates from each CV model are highly correlated, and thus, their mean can have high variance.
- A better choice is a k-fold CV with $k = 5$ or $k = 10$.
- It is often used in modeling because it has been empirically demonstrated to yield results that do not have too much bias or variance.

4.4 K-Fold Cross-Validation

Cross-Validation on Classification Problems

- Here we use the number of misclassified observations (instead of the MSE) to quantify test error.
- The LOOCV error rate takes the form:

$$CV_n = \frac{1}{n} \sum_{i=1}^n Err_i$$

where $Err_i = I(Y_i \neq \hat{Y}_i)$

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4.5 Bootstrap

- The bootstrap can be used in various modeling frameworks to estimate the uncertainty associated with a given estimator.
- For example, bootstrap is useful to estimate the standard errors of the coefficients for a linear regression.
- The bootstrap involves repeated sampling with replacement from the original dataset to form a distribution of the statistic in question.

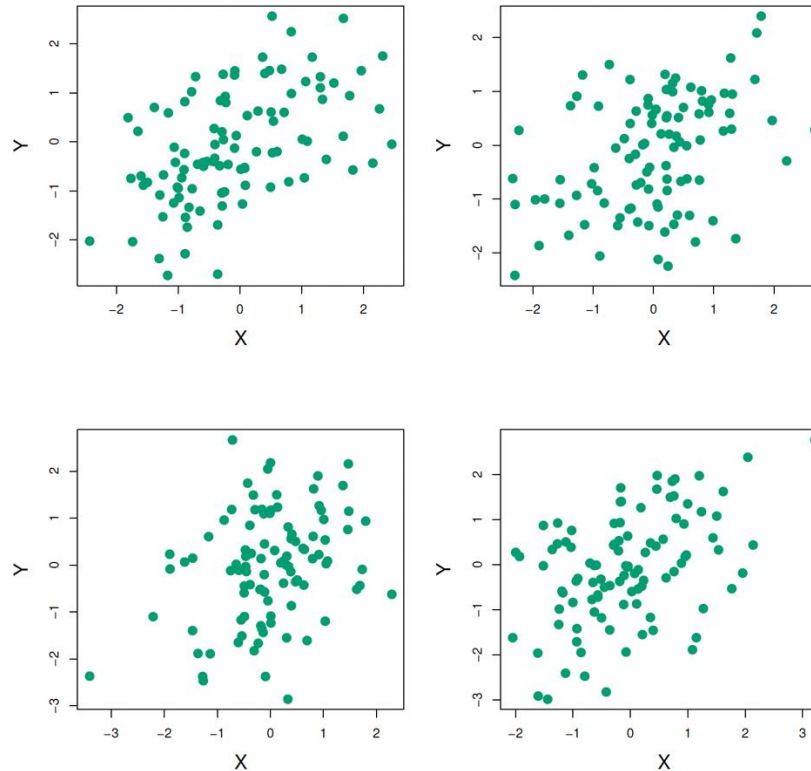
4.5 Bootstrap

A simple example

- Suppose we wish to invest a fixed sum of money in two financial assets that yield returns of X and Y , respectively, where X and Y are random quantities.
- We will invest a fraction α of our money in X , and will invest the remaining $1 - \alpha$ in Y .
- We wish to choose α to our investment's total risk, or variance.
- We can estimate them from a dataset that contains measurements for X and Y .

4.5 Bootstrap

A simple example – continued



Simulated 100 pairs of data points (X, Y) four times and got four values for α ranging from 0.532 to 0.657. From left to right and top to bottom, the resulting estimates for α are 0.576, 0.532, 0.657, and 0.651.

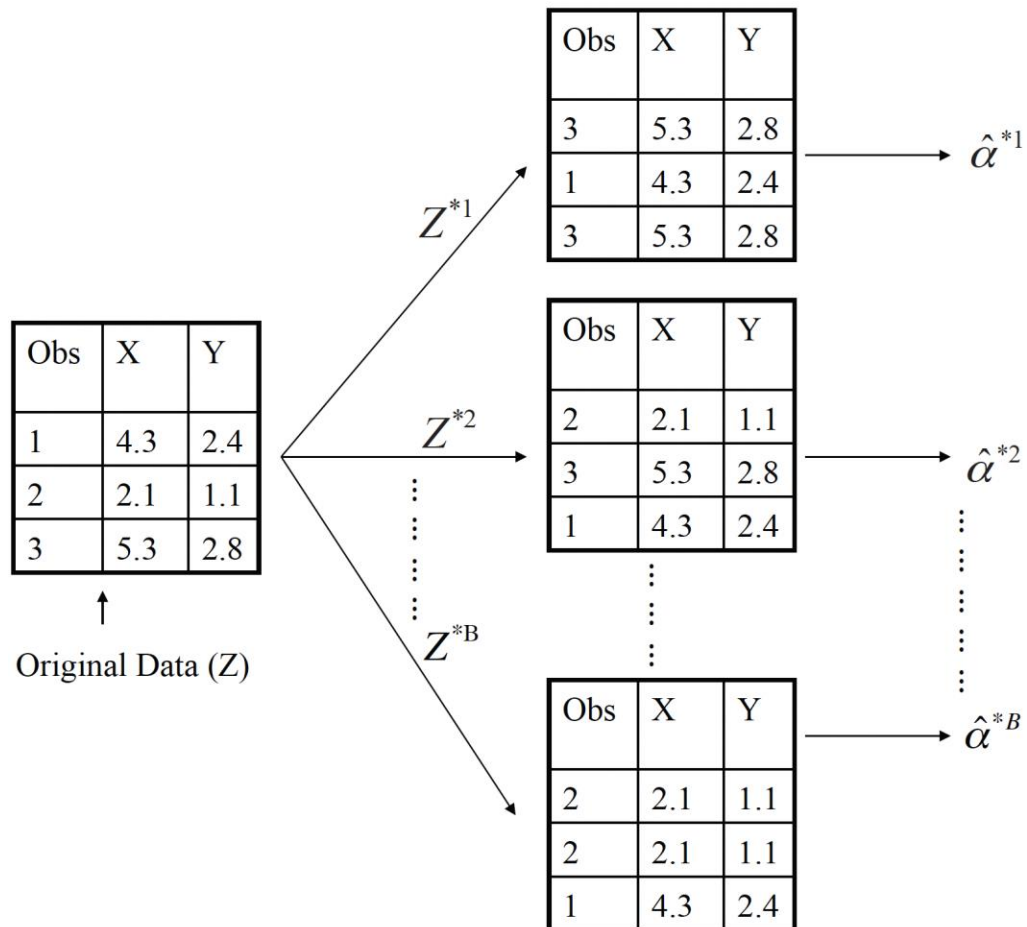
4.5 Bootstrap

Real-world scenario

- The procedure outlined above cannot be applied because, for real data, we cannot generate new samples from the original population.
- However, the bootstrap approach allows us to mimic the process of obtaining new datasets to estimate the variability of our estimate without generating additional samples.
- We obtain distinct datasets by **repeatedly** sampling observations from the original dataset **with replacement**.
- Each of these “bootstrap datasets” is created by sampling **with replacement** and is the same size as our original dataset.
- As a result, some observations may appear more than once in a given bootstrap dataset, and some not at all.

4.5 Bootstrap

Real-world scenario – continued



- A graphical illustration of the bootstrap approach on a small sample containing $n = 3$ observations.
- Each bootstrap dataset contains n observations, sampled with replacement from the original data set.
- Each bootstrap dataset is used to obtain an estimate of α .

4.5 Bootstrap

Real-world scenario – continued

- In more complex data situations, figuring out the appropriate way to generate bootstrap samples can require some thought.
- For example, if the data is a time series, we can't simply sample the observations with replacement (*why not?*).
- Instead, we can create blocks of consecutive observations and sample those with replacements.
- Then, we paste the sampled blocks together to obtain a bootstrap dataset.

4.5 Bootstrap

Can the bootstrap estimate prediction error?

- In cross-validation, each of the K validation folds is distinct from the other $K - 1$ folds used for training: there is no overlap. This is crucial for its success. Why?
- To estimate prediction error using the bootstrap, we could consider using each bootstrap dataset as our training sample and the original sample as our validation sample.
- However, each bootstrap sample has a significant overlap with the original data. About two-thirds of the original data points appear in each bootstrap sample.
- This will cause the bootstrap to underestimate the true prediction error seriously. Why?
- The other way around – with original sample = training sample, bootstrap dataset = validation sample – is worse!

4.5 Bootstrap

Removing the Overlap

- We can partly fix this problem by only using predictions for those observations that did not (by chance) occur in the current bootstrap sample.
- However, the method gets complicated, and in the end, cross-validation provides a simpler, more attractive approach for estimating prediction errors.

Summary

- We have two main resampling methods: cross-validation and the bootstrap. Both refit a model to samples from the training set to obtain additional information about the fitted model.
- For example, they provide estimates of test-set prediction error and the standard deviation and bias of parameter estimates.
- The training error rate is often quite different from the test error rate, and it can be dramatically underestimated.
- The best solution is to have a large test set (which is not often available).

Further Reading

- Chapter 5 – James G., Witten D., Hastie T., Tibshirani R., & Taylor J. (2023) *An Introduction to Statistical Learning with Applications in Python*. Springer
<https://link.springer.com/book/10.1007/978-3-031-38747-0>
- Chapter 5 - Introduction to Statistical Learning Using R Book Club: <https://r4ds.github.io/bookclub-islr/resampling-methods.html>
- Summary of Chapter 5:
<https://www.bijenpatel.com/guide/islr/resampling-methods/>

End of Topic

Thank you
Any questions?